

Numerical Analysis - I

Course Name :- Numerical Analysis - I

Course Code :- MAT 1202.

Assessment Strategy.

• End Semester Exam = 70%

- Theory - 50%

- Practical - 20%

• Home Assignments - 15%

• Mid Semester Exam - 15%

Chapter 01 - Number System

Chapter 02 - Roots of non linear equations

Chapter 03 - Interpolation and polynomial approximation.

Chapter 04 - Numerical Differentiation and Integration
use of MATLAB.

Chapter 01:- Number Systems!!!

most commonly used numbering system is decimal number system. This system as its name indicates is based on the number 10.

The number 10 is called the basis or radix of the system.

To indicate a number in the system we use the symbols: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. These symbols are called digits.

When using numbers in some other base we will use $(x)_\beta$ to indicate that number x is to be interpreted in the β number system.

• The binary number system

This system is based on the number 2. The digits are 0 and 1 $\beta = 2$.

Ex:- $(1101)_2$, $(1111)_2$, $(11011.011)_2$.

• The Octal System:-

This system is based on the number 8 ($=\beta$) digits are :- 0, 1, 2, 3, 4, 5, 6, 7

Ex:- $(503)_8$, $(1237)_8$, $(5413.706)_8$.

Hexadecimal System:

This system is based on 16 ($= \beta$)

Digits are :- 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

Here, $A = 10$, $B = 11$

The reason for using letters instead of ^{Combination of} using numerical symbols to represent symbols higher than 9 to keep the all the symbols.

ex:- $(56C.F)_{16}$, $(1A5F)_{16}$.

Let β be a positive integer.

Then β different symbols (digits) are used to write a number.

Converting Decimal to other bases.

01). Decimal to binary.

Let x be a positive decimal integer.

Let $x = (a_n a_{n-1} a_{n-2} \dots a_1 a_0)_2$.

$$\Rightarrow x = a_n \times 2^n + a_{n-1} \times 2^{n-1} + a_{n-2} \times 2^{n-2} + \dots + a_1 \times 2^1 + a_0 \times 2^0$$

$$x_0 = \frac{x}{2} = \frac{a_n \times 2^{n-1} + a_{n-1} \times 2^{n-2} + a_{n-2} \times 2^{n-3} + \dots + a_1 \times 2^0 + \underbrace{a_0}_{\text{Remind}}}{2}$$

$x_1 = \text{quotient.}$

$$\frac{x_1}{2} = \frac{a_n \times 2^{n-2} + a_{n-1} \times 2^{n-3} + a_{n-2} \times 2^{n-4} + \dots + a_2 \times 2^0 + \underbrace{a_1}_{\text{Remind}}}{2} = x_2$$

$$\frac{x_2}{2} = a_n \times 2^{n-3} + a_{n-1} \times 2^{n-4} + a_{n-2} \times 2^{n-5} + \dots + a_3 \times 2^1 + \frac{a_2}{2} \Rightarrow a_2$$

$$\frac{x_{n-1}}{2} = \underbrace{a_n}_{x_n} + \frac{a_{n-1}}{2}$$

$$\frac{x_n}{2} = \frac{a_n}{2}$$

Eg:- $5 = (a_2, a_1, a_0)_2$

$$5 = a_2 \times 2^2 + a_1 \times 2^1 + a_0 \times 2^0$$

$$\frac{5}{2} = \underbrace{a_2 \times 2^1 + a_1 \times 2^0}_{x_1} + \frac{a_0}{2} = 2 + \frac{1}{2} \Rightarrow a_0 = 1$$

$$2 = a_2 \times 2^1 + a_1 \times 2^0$$

$$\frac{2}{2} = a_2 \times 2^0 + \frac{a_1}{2} = 1 + \frac{0}{2} \Rightarrow a_1 = 0, a_2 = 1$$

Now it is Clear that a_i is the remainder when x_i is divided by 2 for all $i = 1, 2, \dots, n$

This procedure is performed until we get the zero quotient.

This procedure ^{may be extended to} ~~is performed~~ Convert any other positive Such as octal, hexadecimal etc.

Eg: what is the binary equivalent of decimal 41?

$$\begin{array}{r}
 2 \overline{)41} \\
 2 \overline{)20} - 1 \\
 2 \overline{)10} - 0 \\
 2 \overline{)5} - 0 \\
 2 \overline{)2} - 1 \\
 2 \overline{)1} - 0 \\
 0 - 1
 \end{array}$$

Remainders.

$$41 = (101001)_2$$

what is the octal equivalent of decimal 41?

$$8 \overline{)41}$$

$$8 \overline{)5} - 1$$

$$41 = (51)_8$$

- How to Convert a positive decimal fraction, $x (< 1)$ to its binary equivalent or its other equivalents?

$$\text{Let } x = (.a_1 a_2 a_3 \dots a_n)_2$$

$$= a_1 \times 2^{-1} + a_2 \times 2^{-2} + a_3 \times 2^{-3} + \dots + a_n \times 2^{-n}$$

$$2x_1 = a_1 + a_2 \times 2^{-1} + a_3 \times 2^{-2} + \dots + a_n \times 2^{-n+1}$$

x_2

$$2x_2 = a_2 + a_3 \times 2^{-1} + \dots + a_n \times 2^{-n+2}$$

x_3

$$2x_n = a_n + a_{n-1} \times 2^{-1} + a_{n-2} \times 2^{-2} + \dots$$

$$x = (a_1 a_2 a_3 \dots a_n)_\beta$$

$$x = a_1 \times \beta^{-1} + a_2 \times \beta^{-2} + \dots + a_n \times \beta^{-n}$$

$$\beta x = a_1 + a_2 \times \beta^{-1} + \dots + a_n \times \beta^{-n+1}$$

Integer part of βx .

$$\beta x_1 = a_1 + a_2 \times \beta^{-1} + \dots + a_n \times \beta^{-n+2}$$

multiply by β to get a_2 .

01. what is binary representation of decimal

$$x = 0.1$$

Converting results:

$$0.1$$

$$0$$

$$0.1 \times 2 = 0.2 = 0 + 0.2 \quad 0.0$$

$$2 \times 0.2 = 0.4 = 0 + 0.4 \quad 0.00$$

$$2 \times 0.4 = 0.8 = 0 + 0.8 \quad 0.000$$

$$2 \times 0.8 = 1.6 = 1 + 0.6 \quad 0.0001$$

$$2 \times 0.6 = 1.2 = 1 + 0.2 \quad 0.00011$$

$$2 \times 0.2 = 0.4 = 0 + 0.4 \quad 0.000110$$

$$2 \times 0.4 = 0.8 = 0 + 0.8 \quad 0.0001100$$

$$2 \times 0.8 = 1.6 = 1 + 0.6 \quad 0.00011001$$

$$2 \times 0.6 = 1.2 = 1 + 0.2 \quad 0.000110011$$

$$2 \times 0.2 = 0.4 = 0 + 0.4 \quad 0.0001100110$$

Floating Point Number Systems

Although quantities and practical problems deal with real valued quantities and theorems upon which numerical methods are based are written in terms of real valued functions.

Computers like measurement devices have more concepts of the real numbers instead computers represent real numbers in what is known as a floating point number systems.

Defⁿ:

A floating point number system $F(p, k, m, M)$ is a subset of the real number system characterised by the parameters.

p - The base

k - The number of digits in the base p expansion

m - The minimum exponent

M - The Maximum exponent

e - exponent.

Elements of $F(p, k, m, M)$ are those real numbers that can be expressed exactly as $\pm (0.d_1 d_2 d_3 \dots d_k)_p \times p^e$,

where

$$0 \leq d_k \leq p-1, \quad m \leq e \leq M$$

Also, note that when $f \neq 0$, $d_1 \neq 0$. But $f = 0$, $d_1 = 0$.

$$(0.d_1 d_2 d_3 \dots d_k)_p = \frac{d_1}{p} + \frac{d_2}{p^2} + \frac{d_3}{p^3} + \dots + \frac{d_k}{p^k}.$$

Alisa

is called the mantissa of the floating point number

The number p is called radix and M is called radix point.

Eg:-

Consider the system $f(10, 2, 0, 2)$

Here $p = 10$, $k = 2$, $m = 0$ and $M = 2$

Also,

$0 \leq e \leq 2$, So $e = 0, 1, 2$

Digits in base p system : $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$
 $f = (0.d_1d_2)_{10} \times 10^e$ if $f \neq 0$

d_1	d_2	d_1d_2
1	0	10, 11, 12, 13, 14, 15, 16, 17, 18, 19
2	1	20, 21, 22, 23, 24, 25, 26, 27, 28, 29
3	2	
4	3	
5	4	
6	5	
7	6	
8	7	
9	8	
	9	

When $e = 0$

$\pm (0.d_1d_2) \times 10^0 = \pm (0.d_1d_2) = \pm (0.10), \pm (0.11), \dots,$
 $\pm (0.90), \pm (0.91), \dots, \pm (0.99)$

The no of elements = $2 \times 90 = 180$

when $l=1$ $\pm (0.d_1d_2) \times 10^1 = \pm d_1d_2 = \pm 1.0, \pm 1.1, \dots, \pm 1.9,$

$\vdots$
 $\pm 9.0, \pm 9.1, \dots, \pm 9.9$

The no of elements = $2 \times 90 = 180$.

when $l=2$.

$\pm (0.d_1d_2) \times 10^2 = \pm d_1d_2 = \pm 10, \pm 11, \dots, \pm 19,$

$\vdots$

$\pm 90, \pm 91, \dots, \pm 99$

The no of elements = $2 \times 90 = 180$.

Also note that $0 \in F(10, 2, 0, 2)$

$\therefore$ Total number of elements = $3 \times 180 + 1$

$$= 540 + 1$$

$$= 541$$

Round off errors :-

Suppose that the floating point number system $F(10, 2, 0, 2)$ is used in a hypothetical Computer.

$\beta = 10$
$k = 2$
$m = 0$
$M = 2$

← Computer.

A given calculation require the value $\sqrt{7.1} \approx 2.66458$

Since, this no has more than 2 digits, it is not an element in $F(10, 2, 0, 2)$

$2.66458 \notin F(10, 2, 0, 2)$

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Now, how can we represent the number in this number system (computer) $(.b, b.0)$?

This may be performed using either of the following

- (i) Dropping all digits after the second digit, take the approximation $\sqrt{7.1} \approx 2.6$. This approach is called chopping the number.
- (ii) Rounding the number to two digits of accuracy take the approximation $\sqrt{7.1} \approx 2.7$. This approach is called rounding the number.

In general, let y be a real number whose expansion is given by,

$$y = \pm (0.d_1 d_2 \dots d_k d_{k+1} d_{k+2} \dots)_\beta \times \beta^e$$

with $d_1 \neq 0$ and $m \leq e \leq M$

Let $f(y)$ denote the floating point equivalent of y . This is $f(y) \in F(\beta, k, m, M)$

when the number is chopped,

$$f|_{\text{cop}}(y) = \pm (0.d_1 d_2 \dots d_k)_\beta \times \beta^e$$

when the number is rounded,

$$f|_{\text{round}}(y) = \begin{cases} \pm (0.d_1 d_2 \dots d_k)_\beta \times \beta^e & \text{if } d_{k+1} < \beta/2 \\ \pm (0.d_1 d_2 \dots d_k)_\beta + \beta^{-k} \times \beta^e & \text{if } d_{k+1} \geq \beta/2 \end{cases}$$

Defⁿ :-

The error introduced by converting real number to its floating equivalent is called Round off error.

Defⁿ :-

Let p^* denote any approximation to the number p the absolute error in p^* is given by

$$|p - p^*|$$

The relative error in p^* is given by

$$\left| \frac{p - p^*}{p} \right|$$

Floating Point Arithmetic

Let x and y be two real numbers. Let $\oplus$, $\ominus$, $\otimes$ and $\oslash$ denote the corresponding floating point equivalent operations of arithmetic operations $+$, $-$, $\times$ and $\div$.

Real Number System

+	$\times$
-	$\div$

Floating number System

$\oplus$	$\otimes$
$\ominus$	$\oslash$

Non floating operations are defined.

$$x \oplus y : = f \mid [f \mid (x) + f \mid (y)]$$

$$x \ominus y : = f \mid [f \mid (x) - f \mid (y)]$$

$$x \otimes y : = f \mid [f \mid (x) \times f \mid (y)]$$

$$x \oslash y : = f \mid [f \mid (x) \div f \mid (y)]$$

Eg:-

Using 4 decimal digits rounding arithmetic find $\frac{5}{3} \otimes \beta$

$$\frac{5}{3} = 1.66667$$

$$\beta = 1.73205$$

$$\frac{5}{3} \otimes \beta = 1.66667 \otimes 1.73205$$

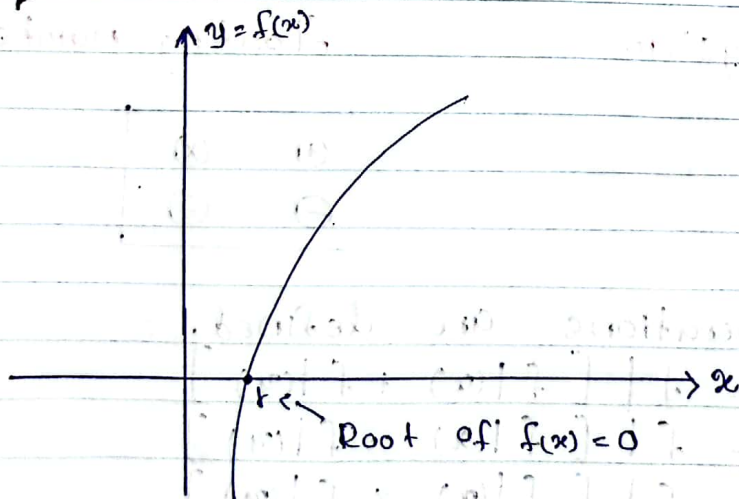
$$= f_{\text{round}} \left[f_{\text{round}}(1.66667) \times f_{\text{round}}(1.73205) \right]$$

$$= f_{\text{round}} [1.667 \times 1.732]$$

$$= f_{\text{round}} (2.887)$$

CHAPTER 02 :- Roots of Non-linear Equations:-

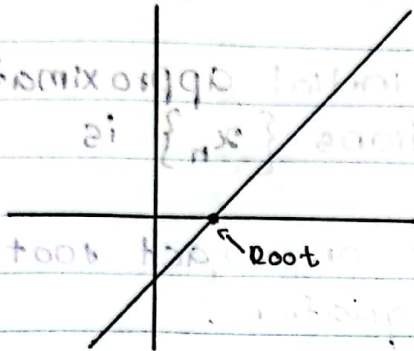
Let $f(x)$ be a function. A number r satisfying $f(r) = 0$ is called a root of the equation $f(x) = 0$



Special Case:-

01) $f(x) = a_1x + a_0$, $a_1 \neq 0$

$-\frac{a_0}{a_1}$ is the root of $f(x) = 0$.

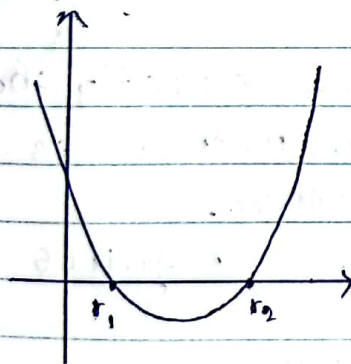


02) $f(x) = a_2x^2 + a_1x + a_0$, $a_2 \neq 0$ There are two roots.

$$r_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2a_0}}{2a_2} \quad r_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2a_0}}{2a_2}$$

Let: $f(x) = x^2 - 5x + 6$

$r_1 = 2$ and $r_2 = 3$ are roots of $f(x) = 0$.



03). Let $f(x) = a_3x^3 + a_2x^2 + a_1x + a_0$

There are formulas.

Let $f(x) = a_nx^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$, $a_n \neq 0$

when $n \geq 5$ there are no formulas to directly find the roots of the equation $f(x) = 0$.

Also, there are no direct methods to find a root of a transcendental equation. Such as $e^x = x$, $\tan x = x$,

$xe^{x-1} = 0$, $\cos x - xe^x = 0$, etc.

We need some approximation method.

Iterative Methods :-

These methods are based on successive approximations for a desired root.

Starting with one or more initial approximations a sequence of approximations $\{x_n\}$ is obtained such that,

$$\lim_{n \rightarrow \infty} x_n = r, \text{ where } r \text{ is an exact root of a desired equation.}$$

An iterative method for finding a root of the equation $f(x) = 0$ takes the form:

$$x_{n+1} = \phi(x_n) \text{ for all } n = 0, 1, 2, 3, \dots$$

Here, x_0 is called the initial approximation. $\phi(x)$ is called the iteration function.

$x_{n+1} = \phi(x_n)$, $n = 0, 1, 2, \dots$, requires exactly one initial value. Such an iteration method is called a one point iteration method.

$x_{n+1} = \phi(x_{n-1}, x_n)$, $n = 0, 1, 2, \dots$, requires two initial values.

Such an iteration method is called a two-point iteration method.

$\epsilon = 0.02$	
x_n	$ x_{n-1} - x_n < 0.02$
x_0	} 0.5
x_1	
x_2	} 0.4
x_3	
x_4	
x_5	} < 0.02
x_6	

Stopping Criterion :-

Since we can not perform infinite number of iterations we have to stop the iteration when

$$|f(x_n)| \leq \epsilon \quad \text{or} \quad |x_{n+1} - x_n| \leq \epsilon$$

Exact value of $f(x) = 0$, where $x = r$, the exact root.
 Error = $|f(x_n) - f(r)| = |f(x_n) - 0|$

$$= |f(x_n)|$$

$$|x_{n+1} - x_n| = |x_{n+1} - r + r - x_n|$$

$$\leq |x_{n+1} - r| + |x_n - r|$$

$$\leq \epsilon$$

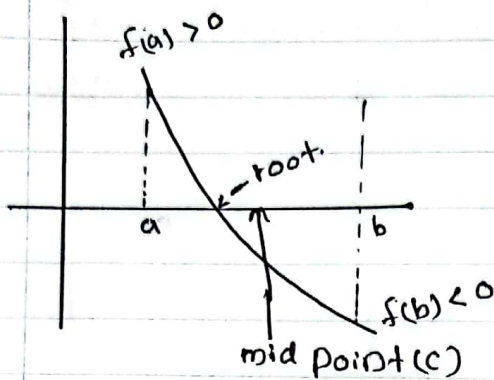
This ϵ has to be given.

ϵ - Tolerance

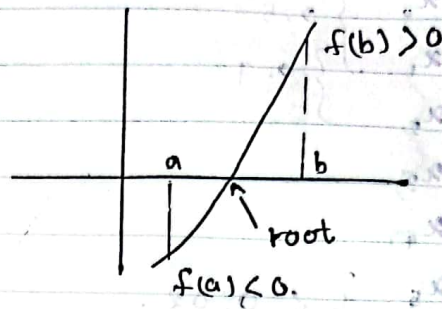
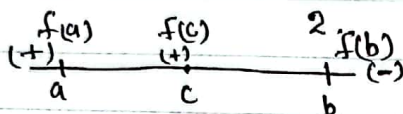
= Tol

Bisection Method :-

$$y = f(x)$$



$$c = \frac{a+b}{2}$$

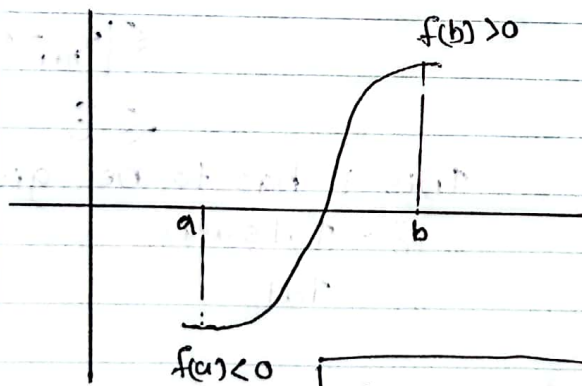


The bisection method is based on the following theorem:-

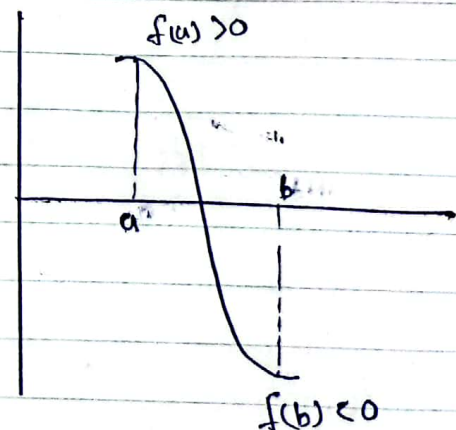
* Intermediate Value Theorem (IVT)

Let f be a continuous function over the closed interval $[a, b]$.

Suppose that $f(a)$ and $f(b)$ have opposite signs. then, there exists $c \in [a, b]$ such that $f(c) = 0$.



$$f(a) \times f(b) < 0$$



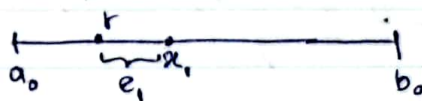
$$f(a) \times f(b) < 0$$

Suppose that a root of $f(x)=0$, lies in the interval $I_0 = (a, b) = (a_0, b_0)$

1. Bisect I_0 at the point $x_1 = \frac{a_0 + b_0}{2}$
 $\delta \approx x_1$
2. If $f(a_0) f(x_1) < 0$ let $I_1 = (a_0, x_1) = (a_1, b_1)$
 If $f(x_1) f(b_0) < 0$ let $I_1 = (x_1, b_0) = (a_1, b_1)$
3. Bisect I_1 at the mid point of I_1
 $x_2 = \text{th mid point of } I_1$
 $\delta \approx x_2$

keep bisecting and selecting an interval including the root until a given ~~accuracy~~ accuracy is met.

$$x_1 = \frac{a_0 + b_0}{2}$$



$$\Rightarrow e_1 = |r - x_1| \leq \frac{1}{2}(b_0 - a_0)$$

$$x_2 = \frac{a_1 + b_1}{2} \Rightarrow e_2 = |r - x_2| \leq \frac{1}{2}(b_1 - a_1)$$

$$x_3 = \frac{a_2 + b_2}{2} \Rightarrow e_3 = |r - x_3| \leq \frac{1}{2}(b_2 - a_2)$$

At the n^{th} iteration:-

$$x_n = \frac{a_{n-1} + b_{n-1}}{2} \Rightarrow e_n = |r - x_n| \leq \frac{1}{2}(b_{n-1} - a_{n-1}) \quad \text{--- ①}$$

for all $n = 1, 2, 3, \dots$

$$b_{n-1} - a_{n-1} = \frac{1}{2}(b_{n-2} - a_{n-2})$$

$$= \frac{1}{2} \cdot \frac{1}{2} (b_{n-3} - a_{n-3})$$

$$\vdots$$

$$= \frac{1}{2^{n-1}} (b_0 - a_0)$$

Now ① $\Rightarrow$ Suppose that a root of $f(x)$ lies in (a_n, b_n) .
 $e_n = |t - x_n| \leq \frac{1}{2} \cdot \frac{1}{2^{n-1}} (b_0 - a_0)$

$$= \frac{1}{2^n} (b_0 - a_0)$$

$$= \frac{1}{2^n} (b - a)$$

Suppose for a given error ϵ ,

$$\frac{b_0 - a_0}{2^n} \leq \epsilon$$

$$\Rightarrow b_0 - a_0 \leq \epsilon \cdot 2^n$$

$$\log(b_0 - a_0) \leq \log(\epsilon \cdot 2^n)$$

$$= \log \epsilon + n \log 2$$

$$\Rightarrow n \geq \frac{\log(b_0 - a_0) - \log \epsilon}{\log 2}$$

$\lceil x \rceil$ = Ceiling function

is an integer such that

$$\lceil x \rceil \geq x$$

$\uparrow$ smallest integer $\geq x$

$$\lceil 1.5 \rceil = 2$$

$$\lceil 1.1 \rceil = 2$$

$$\lceil 3 \rceil = 3$$

$\lfloor x \rfloor$ = floor function.

an integer such that

$$\lfloor x \rfloor \leq x$$

Largest integer $\leq x$

$$\lfloor 1.3 \rfloor = 1$$

$$\lfloor -2.3 \rfloor = -3$$

$$\lfloor 5 \rfloor = 5$$

$$\text{Set } n = \left\lceil \frac{\log(b_0 - a_0) - \log \epsilon}{\log 2} \right\rceil$$

Let $\epsilon = 10^{-2} = 0.01$ and Let $a_0 = 0, b_0 = 1$

$$\frac{\log(b_0 - a_0) - \log \epsilon}{\log 2} = \frac{\log(1-0) - \log(10^{-2})}{\log 2} = 6.6438561$$

$$\therefore n = \lceil 6.6438561 \rceil = \underline{\underline{7}}$$

ϵ	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}
n	7	10	14	17	20	24

example:-

Consider the equation.

$$x^3 - 2x - 1 = 0$$

Show that this eqⁿ has a root between 1 and 2.

Find it correct to 4 decimal places.

$$\text{Let } f(x) = x^3 - 2x - 1$$

$$\begin{array}{ccc} (-) & (-) & (+) \\ \hline 1 & & 2 \end{array}$$

$$f(1) = 1^3 - 2(1) - 1 = -2 < 0$$

$$f(2) = 2^3 - 2(2) - 1 = 3 > 0$$

$$\therefore f(1) f(2) < 0$$

There must be a root of between 1 and 2.

$$x_1 = \frac{1+2}{2} = 1.5$$

$$f(x_1) = (1.5)^3 - 2(1.5) - 1 < 0$$

Since $f(x_1) f(2) < 0$, the root must be between x_1 and 2

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$$f = g(x) = x^2 \exp(x) - 1$$

$$a = 1/2, b = 3/2, x_0 = a$$

$$x_2 = \frac{a+b}{2} = \frac{1.5+2}{2} = 1.75$$

$$h=1$$

$$E = 0.0005$$

$$Err = abs(x_0 - a)$$

$$f(x_2) = f(1.75) = (1.75)^3 - 2(1.75) - 1 > 0$$

Since $f(x_1) \cdot f(x_2) = f(1.5) \cdot f(1.75) < 0$, the root must be between x_1 and x_2 .

We need to calculate the root correct to 2 decimal place.

Note:-

Let x be a number and x^* be an approximation for x . If $|x - x^*| \leq 0.5 \times 10^{-n}$

then x^* represents x accurate to n decimal places (n significant digits).

$$\text{Let } E = 0.5 \times 10^{-4} = 0.00005$$

$$|x_{n+1} - x_n| \leq E$$

$$|x_2 - x_1| = |1.75 - 1.5| = 0.25 > 0.00005$$

$$|x_3 - x_2| = 0.125 > 0.00005$$

the.

From a MATLAB Program, we have

$$|x_{15} - x_{14}| = 0.00003 < 0.00005.$$

homework! Show that $x^2 - x = 0$ has a root between $1/2$ and $3/2$ and determine it.

Show that $x^2 - 1 = 0$ has a root between

0.5 and 1.0. Determine it correct to 3 decimal places.

Atlas

MATLAB built in functions for finding roots.

The `fzero` Command Can be used to solve non-linear equations. This needs to know approximately where the solution (root) is,

$$x = \text{fzero}(f(x), x_0)$$

$\uparrow$ Solution $\uparrow$ non-linear function $\nwarrow$ Approximate Solution.

If $x = \cos x$

$x \rightarrow \cos x$,

the $f(x) = x - \cos x$

$$f(x) = x - \cos x$$

$$x_0 = 2 \quad f = @(x) x - \cos(x);$$

$$x = \text{fzero}(f, 1);$$

$f(x) = 0$ — we want to solve this.

root Command:-

This can be used to find a root of polynomial equation $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$
 a_n n^{th} degree polynomial

In MATLAB:-

$$P = [a_n \quad a_{n-1} \quad a_{n-2} \quad \dots \quad a_1 \quad a_0]$$

$$P(x) = x^3 - 2x - 1$$

$$P = [1 \quad 0 \quad -2 \quad -1];$$

$$\text{roots}(P)$$

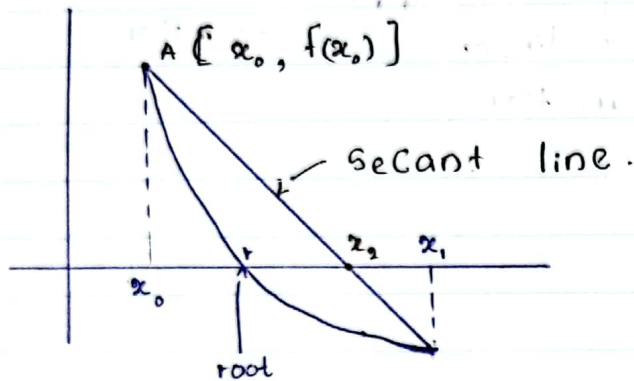
Regula-falsi Method / Method of false position.

Let x_0 and x_1 be two initial approximations to the root r of the equation $f(x) = 0$.

We also assume that the root lies between

x_0 and x_1 ,

So, $f(x_0) f(x_1) < 0$.



x_2 : The next approximation to the root r .

This is the point at which the secant line through the points A and B cuts x -axis.

The equation of the secant line AB is,

$$\frac{y - f(x_0)}{x - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$y = \left[\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right] (x - x_0) + f(x_0) \quad \text{--- (1)}$$

Next we want to find x_2 . To get this we

substitute $x = x_2$ and $y = 0$ in (1)

$$\therefore 0 = \left[\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right] (x_2 - x_0) + f(x_0)$$

$$\therefore x_2 - x_1 = -f(x_1) \left[\frac{x_1 - x_0}{f(x_1) - f(x_0)} \right]$$

$$x_2 = x_1 - f(x_1) \left[\frac{x_1 - x_0}{f(x_1) - f(x_0)} \right]$$

$$\frac{y - f(x_1)}{x - x_1} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$y = \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x - x_1) + f(x_1) \quad \text{--- ①}$$

Next we want to find x_2 . To get this we substitute $x = x_2$ and $y = 0$ in ①

$$0 = \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x_2 - x_1) + f(x_1)$$

$$\therefore (x_2 - x_1) = -f(x_1) \frac{(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = x_1 - f(x_1) \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

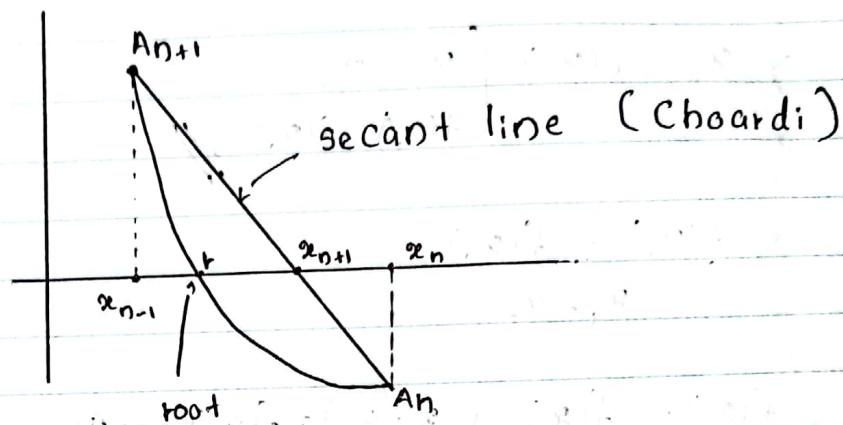
$$x_0 = x_1 - T_0$$

$$x_0 = x_2 - T_1$$

$$x_3 = x_2 - \frac{(x_2 - x_1)}{f(x_2) - f(x_1)} \cdot f(x_2)$$

$$x_4 = x_3 - \frac{(x_3 - x_2)}{f(x_3) - f(x_2)} \cdot f(x_3)$$

Suppose that x_{n-1} and x_n are two approximation to the root r of $f(x) = 0$.



Then in general we have,

$$x_{n+1} = x_n - \frac{(x_n - x_{n-1}) f(x_n)}{f(x_n) - f(x_{n-1})}$$

for all $n = 0, 1, 2, 3, 4, \dots$

In this method at each step, we have to find the interval that includes the root.

Example:-

Consider $f(x) = x^3 - 5x + 1 = 0$

- (i) Show that $f(x) = 0$ has a root between 0 and 1
- (ii) Determine it using RFM.

(i)

Set $x_0 = 0$ and $x_1 = 1$

$$f(x_0) = f(0) = 1 > 0$$

$$f(1) = f(1) = 1 - 5 \times 1 + 1 = -3 < 0$$

In fact, $-y = f(x)$ is a continuous function,
 $\therefore$ there exists a root of $f(x) = 0$ between
 x_0 and x_1 .

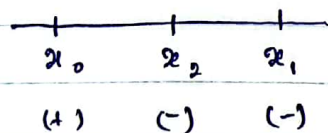
(ii) To determine it, we use RFM.

$$x_2 = x_1 - \frac{(x_1 - x_0) f(x_1)}{f(x_1) - f(x_0)}$$

$$x_2 = 1 - \frac{(1 - 0)(-3)}{-3 - 1}$$

$$\underline{\underline{x_2 = 0.25}}$$

$$f(x_2) = -0.2344 < 0$$

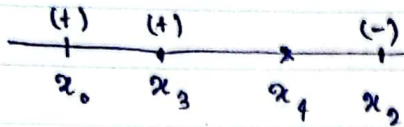


$\therefore$ The root exists between 0 and 0.25.

$$\therefore x_3 = x_2 - \left[\frac{x_2 - x_1}{f(x_2) - f(x_1)} \right] f(x_2)$$

$$= 0.25 - \left[\frac{0.25 - 0}{-0.2344 - 1} \right] (1 - 0.2344)$$

$$\underline{\underline{x_3 = 0.20253}}$$



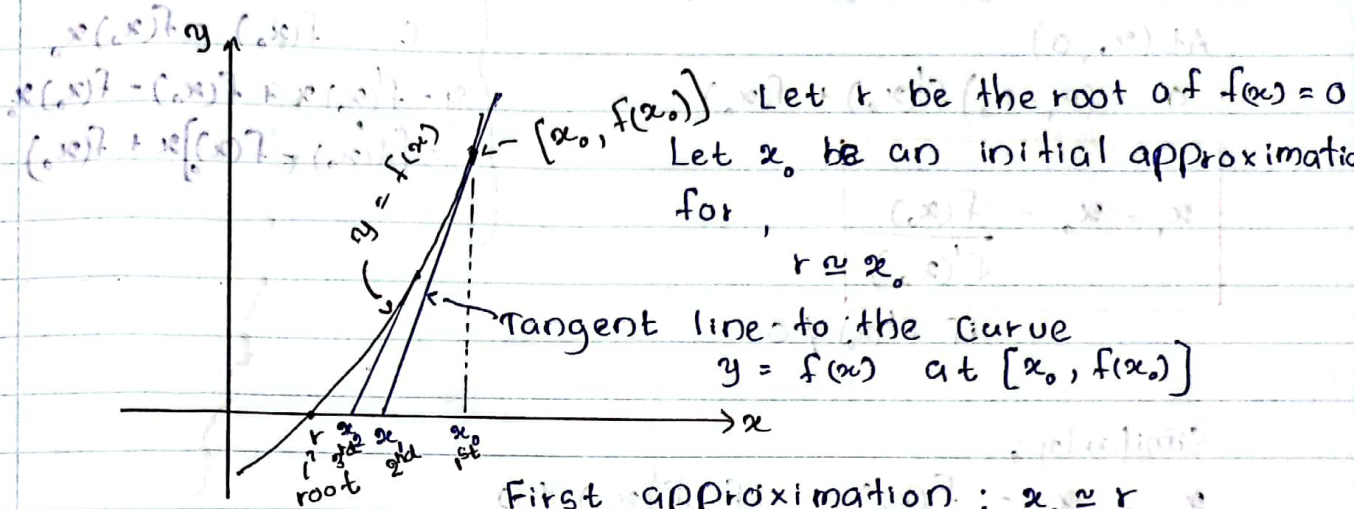
$$x_4 = x_3 - \left(\frac{(x_3 - x_2)}{f(x_3) - f(x_2)} \right) f(x_3)$$

$$x_4 = 0.20253 - \left[\frac{0.20253 - 0.25}{-0.00434 - 0.2344} \right] (-0.00434)$$

$$\underline{\underline{x_4 = 0.2034}}$$

Newton-Raphson Method / Newton Method.

Let f be a function which is continuous and differentiable. $f'(x)$ exists.



The point at which the tangent line to $y = f(x)$ at $[x_0, f(x_0)]$ cuts the x axis

Third approximation: x_3

The point at which the tangent line to $y = f(x)$ at $[x_1, f(x_1)]$ cuts the x axis.

n^{th} approximation: x_{n-1}

The point at which the tangent line to $y = f(x)$ at $[x_{n-2}, f(x_{n-2})]$ cuts the x axis.

$f'(x_0)$: The slope of the tangent line to $y = f(x)$ at $[x_0, f(x_0)]$

The equation of the tangent line to $y = f(x)$ at $[x_0, f(x_0)]$

$$y = (x - x_0) f'(x_0) + f(x_0)$$

At $(x_1, 0)$

$$0 = (x_1 - x_0) f'(x_0) + f(x_0)$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

if $f'(x_0) \neq 0$

$$\begin{cases} y = mx + c \\ y = f(x_0)x + c \\ f(x_0) = f(x_0)x_0 + c \\ c = f(x_0) - f(x_0)x_0 \\ y = f'(x_0)x + f(x_0) - f(x_0)x_0 \\ = [f'(x_0) - f(x_0)]x + f(x_0) \end{cases}$$

Similarly;

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}, \text{ if } f'(x_1) \neq 0$$

At n^{th} approximation, we have,

$$x_{n-1} = x_{n-2} - \frac{f(x_{n-2})}{f'(x_{n-2})}$$

OR

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$$

↑
n+1 approximation.

for $n = 1, 2, 3, \dots$

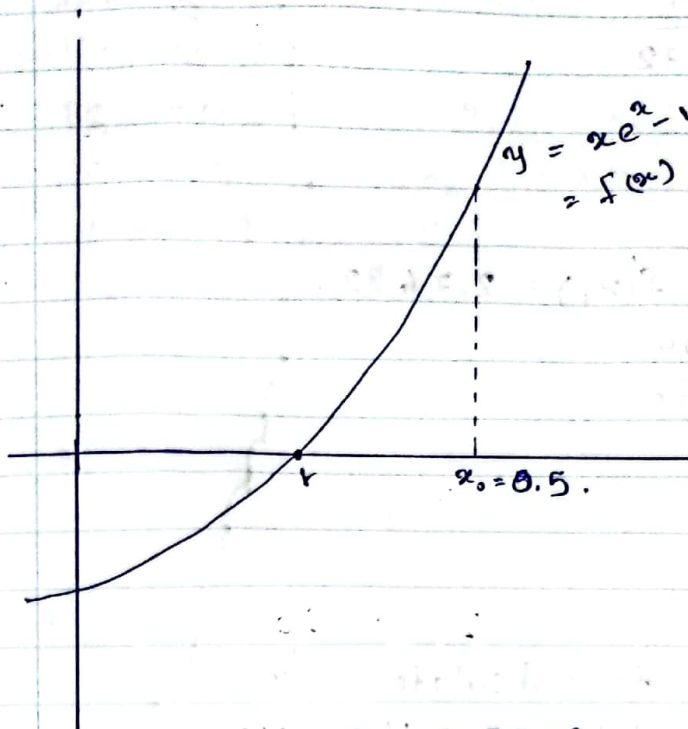
Stopping Criterion:-

(i) Relative error :-

$$\left| \frac{x_n - x_{n-1}}{x_{n-1}} \right| \leq \epsilon = \text{TOL}$$

$$(ii) |f(x_{n-1})| \leq \epsilon = \text{TOL}$$

Example 01.

Consider the eqⁿ : $xe^x - 1 = 0$.Let $x_0 = 0.5$ 

$$f'(x) = xe^x + e^x = (1+x)e^x$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_0 = 0.5$$

$$f(x_0) = f(0.5) = 0.5e^{0.5} - 1$$

$$f'(x_0) = (1+0.5)e^{0.5} = 2.4731$$

$$x_1 = 0.5710, \quad f(x_1) = 0.0107$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f'(x_1) = 2.7808$$

$$= 0.5710 - \frac{0.0107}{2.7808}$$

$$x_2 = 0.5671$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$f(x_2) = 3.3928$$

$$f'(x_2) = 2.7632$$

$$x_3 = 0.5671 - \frac{3.3928}{2.7632}$$

$$x_3 = 0.56715$$

Mathlab code
 $x3 = x2 - f(x2)/g(x2)$
 $g(x2) = f'(x2)$

$$f(x_3) = 3.4119$$

$$f'(x_3) = 2.7632.$$

$$x_4 = 0.5671 - \frac{3.4119}{2.7632}$$

$$\underline{x_4 = 0.56719}$$

$$f(x_4) = 2.2204$$

$$f'(x_4) = 2.7632.$$

$$x_5 = 0.5671 - \frac{2.2204}{2.7632}$$

$$\underline{x_5 = 0.56714}$$

* Secant Method :-

In NR, we have to calculate $f'(x)$

This is a major difficulty of the NR

To overcome it we use,

$$f'(x) = \frac{f(x_{n-1}) - f(x_n)}{x_{n-1} - x_n}$$

Then the NR formula becomes,

$$x_{n+1} = x_n - \frac{f(x_n)(x_{n-1} - x_n)}{f(x_{n-1}) - f(x_n)}$$

for $n = 1, 2, 3, \dots$

we need two initial approximations for the root
Here we do not consider the interval where
the root exists. at each step,

Consider the last problem

$$xe^{x-1} = 0 \quad \text{a root of } xe^{x-1} = 0$$

Set $x_0 = 5$ and $x_1 = 1$

$$x_{n+1} = x_n - \frac{f(x_n)(x_{n-1} - x_n)}{f(x_{n-1}) - f(x_n)}$$

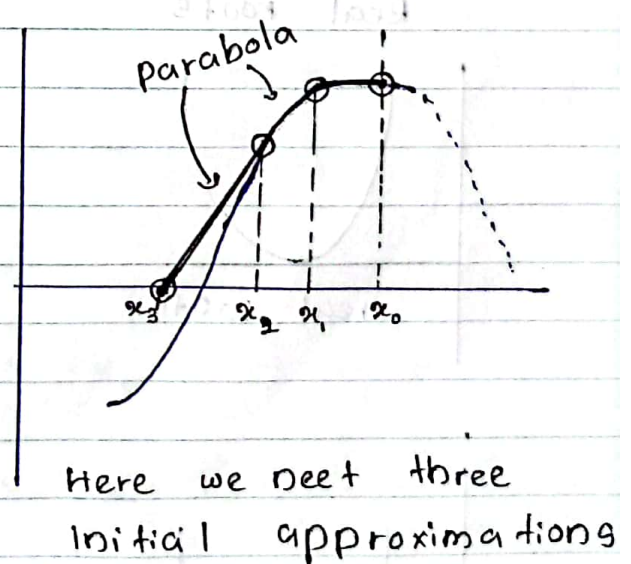
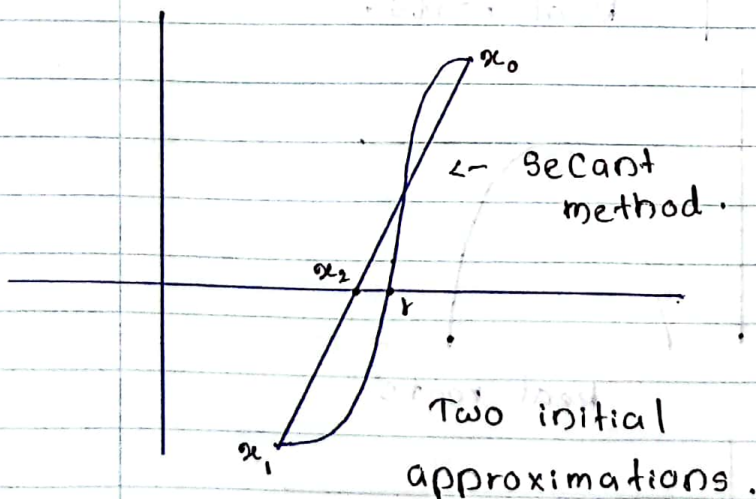
$$x_2 = x_1 - \frac{f(x_1)(x_0 - x_1)}{f(x_0) - f(x_1)}$$

$$x_3 = x_2 - \frac{f(x_2)(x_1 - x_2)}{f(x_1) - f(x_2)}$$

$$x_4 = x_3 - \frac{f(x_3)(x_2 - x_3)}{f(x_2) - f(x_3)}$$

* Müller method :-

The Müller method is an extension of the Secant method.



The Müller method can be used to approximate a root of a real function

Here we start with a real ~~start~~ initial approximations to find a Complex root.

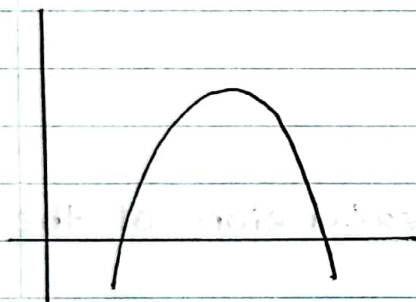
Initial approximations.

$$x_0, x_1, x_2$$

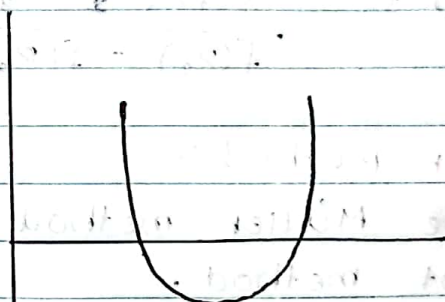
x_3 : new approximation is the point at which the parabola through (x_0, y_0) , (x_1, y_1) and (x_2, y_2) cuts the axis.

Step 01:-

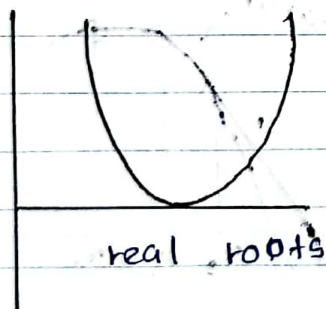
Find the parabola passing through the points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$



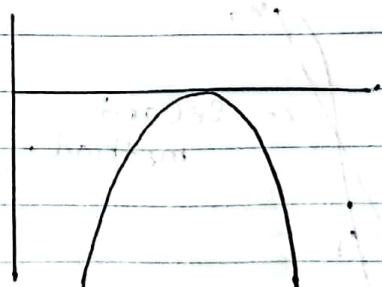
Real roots



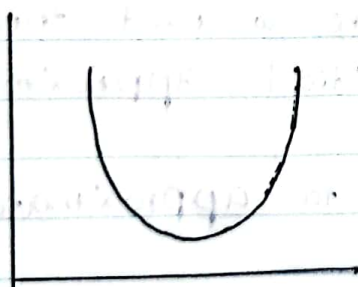
Real roots.



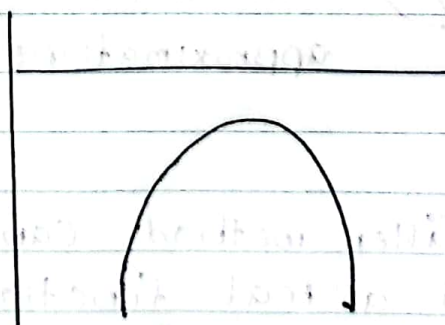
real roots



Real roots



Complex root



Complex roots.

Atlas

Suppose that the equation of the Parabola is given by,

$$y(x) = a(x-x_2)^2 + b(x-x_2) + C = Ax^2 + Bx + C$$

We must determine a , b and C using the points $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$

At $(x_0, f(x_0)) = (x_0, y_0)$

$$y_0 = a(x_0 - x_2)^2 + b(x_0 - x_2) + C = f(x_0) \quad \text{--- ①}$$

At $(x_1, f(x_1)) = (x_1, y_1)$

$$y_1 = a(x_1 - x_2)^2 + b(x_1 - x_2) + C = f(x_1) \quad \text{--- ②}$$

At $(x_2, f(x_2)) = (x_2, y_2)$

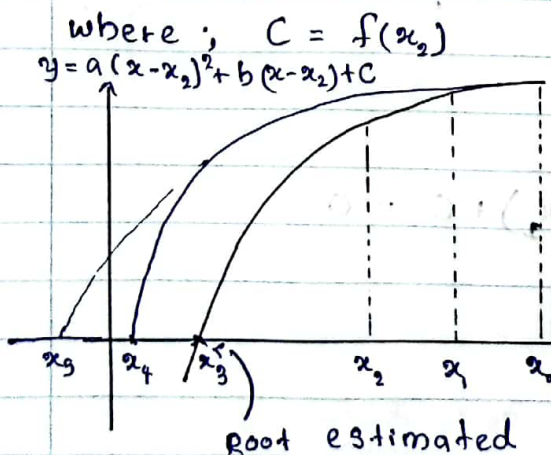
$$y_2 = C = f(x_2) \quad \text{--- ③}$$

From ① and ②, we get

$$a(x_0 - x_2)^2 + b(x_0 - x_2) = f(x_0) - C$$

$$a(x_1 - x_2)^2 + b(x_1 - x_2) = f(x_1) - C$$

$$\begin{bmatrix} (x_0 - x_2)^2 & (x_0 - x_2) \\ (x_1 - x_2)^2 & (x_1 - x_2) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} f(x_0) - C \\ f(x_1) - C \end{bmatrix}$$



① $x_0, x_1, x_2 \Rightarrow x_3$

② $x_1, x_2, x_3 \Rightarrow x_4$

③ $x_2, x_3, x_4 \Rightarrow x_5$

④ $x_3, x_4, x_5 \Rightarrow x_6$

⋮

⑦ $x_{n-1}, x_n, x_{n+1} \Rightarrow x_{n+2}$

Note:-

Consider the quadratic equation

$$ax^2 + bx + c = 0 \quad (*)$$

Set $x = \frac{1}{t}$, where $t \neq 0$

Then we have from (*)

$$a\left(\frac{1}{t}\right) + b\left(\frac{1}{t}\right) + c = 0$$

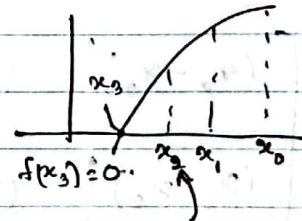
$$ct^2 + bt + a = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2c}$$

$$\therefore \frac{1}{x} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2c}$$

$$\Rightarrow x = \frac{2c}{-b \pm \sqrt{b^2 - 4ac}}$$

$$x = \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}$$



Now, Consider the quadratic equation $y(x_3) = 0$

Solving this gives $x_3 = 0$.

$$y(x_3) = 0 \Rightarrow a(x_3 - x_2)^2 + b(x_3 - x_2) + c = 0$$

$$\Rightarrow x_3 - x_2 = \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}$$

$$\Rightarrow x_3 = x_2 + \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}$$

There are two choices for x_3

$$\text{If } |b + \sqrt{b^2 - 4ac}| > |b - \sqrt{b^2 - 4ac}|$$

$$x_3 = x_2 + \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$

$$\text{If } |b - \sqrt{b^2 - 4ac}| > |b + \sqrt{b^2 - 4ac}|$$

then,

$$x_3 = x_2 + \frac{-2c}{b - \sqrt{b^2 - 4ac}}$$

~~Exap~~ Example 01:-

Approximate a real root of ~~$x^3 - 7x^2 + 6x + 5 = 0$~~ use $x_0 = 0$, $x_1 = 1$ and $x_2 = 2$ as initial approximations.

$$\text{Let } f(x) = x^3 - 7x^2 + 6x + 5$$

Iteration 1: Computing x_3 from x_0 , x_1 and x_2

$$y = a(x - x_2)^2 + b(x - x_2) + c \quad \text{we must determine } a, b, c$$

$$c = f(x_2) = f(2) = 2^3 - 7(2)^2 + 6(2) + 5 = -3$$

$$f(x_0) = f(0) = 5$$

$$f(x_1) = f(1) = 5$$

$$\begin{bmatrix} (x_0 - x_2)^2 & (x_0 - x_2) \\ (x_1 - x_2)^2 & (x_1 - x_2) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} f(x_0) - c \\ f(x_1) - c \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} 4a - 2b \\ a - b \end{bmatrix} = \begin{bmatrix} 8 \\ 8 \end{bmatrix}$$

$$4a - 2b = 8 \quad \text{--- (1)}$$

$$a - b = 8 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \times 4 \Rightarrow 4a - 2b - 4a + 4b = 8 - 32$$

$$2b = -24$$

$$\boxed{b = -12}$$

$$\frac{32}{24}$$

$$a = 8 - 12$$

$$\boxed{a = -4}$$

No: _____

$$Ax = B$$

MATLAB

$$x = A \setminus B$$

A \ B

Date: ____/____/____

$$\textcircled{1} = b \pm \sqrt{b^2 - 4ac}$$

$$= -12 \pm \sqrt{(-2)^2 - 4 \times (-4) \times (-3)}$$

$$= -12 \pm \sqrt{96} \begin{cases} + (-2, 2020) & |-2.2020| \\ - (-21.7980) & |-21.7980| \end{cases}$$

↑
Selected.

$$x_3 = x_2 - \frac{2C}{b - \sqrt{b^2 - 4ac}}$$

$$= 2 - \frac{2(-3)}{-21.7980}$$

$$x_3 = 1.7247$$

Iteration 2:

Here we consider the parabola passing through $(x_1, f(x_1))$, $(x_2, f(x_2))$ and $(x_3, f(x_3))$.

$$x_0, x_1, x_2 \Rightarrow x_3$$

$$x_1, x_2, x_3 \Rightarrow x_4$$

$$C = f(x_3) = f(1.7247) = -0.3437$$

$$f(x_0) = f(x_1) = f(1) = 5$$

$$f(x_1) = f(x_2) = f(2) = -3$$

$$f(1.7247) =$$

$$\begin{bmatrix} (x_1 - x_2)^2 & (x_1 - x_2) \\ (x_2 - x_3)^2 & (x_2 - x_3) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} f(x_1) - c \\ f(x_2) - c \end{bmatrix}$$

$$\begin{bmatrix} 0.5252 & -0.7247 \\ 0.0758 & 0.2753 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5.3437 \\ -2.6563 \end{bmatrix}$$

$$a = -2.2753$$

$$b = -9.0225$$

$$b - \sqrt{b^2 - 4ac} = -17.8700$$

$$x_4 = x_3 - \frac{2c}{b - \sqrt{b^2 - 4ac}}$$

$$b + \sqrt{b^2 - 4ac} = -0.1750$$

$$= 1.7247 - \frac{2(-0.3437)}{-17.8700}$$

$$\underline{\underline{x_4 = 1.6862}}$$

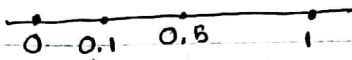
Home work: Approximate a pair of complex root of $x^3 - 2x^2 - 5 = 0$ starting with initial approximation as $x_0 = -1$, $x_1 = 0$ and $x_2 = 1$

Chapter 03 :- Interpolation..

Problem type 01:-

Let $y = f(x)$ be a known function defined the interval $[a, b]$. As we know the function, we can find the value of f at any desired point $x_0 \in [a, b]$.

Ex:- Let $f(x) = x^3 + 1$ on $[0, 1]$. $D(f) \equiv$ domain of $f = [0, 1]$



$$f(0.1) = (0.1)^3 + 1 = 1.001$$

$$f(0.3) = (0.3)^3 + 1 = 1.027$$

$$f(0.5) = (0.5)^3 + 1 = 1.125$$

$$f(0.9) = (0.9)^3 + 1 = 1.729$$

$$f(0.975) = (0.975)^3 + 1 = 1.926859375.$$

Problem type 02:-

Suppose that the following data points giving some values for x and $y = f(x)$

x	0	0.2	0.4	0.6	0.8	1.0
y	7.00	6.008	5.064	4.216	3.512	3.0.

~~The function~~

In this problem we do not know the function that gives the y values in the above data table.

we know only the values of the function f at some given points.

Also

For eg:-

$$f(0) = 7.00$$

$$f(0.2) = 6.008, \text{ etc.}$$

Can we find $f(0.1)$, $f(0.12)$, $f(0.30)$, $f(0.85)$, etc?

In general, we can not find the exact values. But we can approximate / estimate a value using Interpolation.

Interpolation:-

Interpolation is a procedure for estimating / approximating a value between known values of data points.

It is done by determining a mathematical formula that gives the exact values at the given points.

Then the formula is used to estimate a value for a data point between the known points.

One of the most useful groups of functions / formulas used to interpolate an unknown function is the group of polynomials.

Here we use a single polynomial to interpolate an unknown function.

Let $P(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ be an n^{th} degree polynomial. (Here $a_n \neq 0$)

Finding a polynomial $P(x)$ such that $P(x_k) = y_k$ is called polynomial interpolation.

Such a polynomial is called an interpolating polynomial.

Theorem 01:-

Let $x_0, x_1, \dots, x_n$ be distinct points and $y_0, y_1, \dots, y_n$ be the corresponding functional values.

Then, there is a unique (exactly one) polynomial $P(x)$ of degree n .

Such that $P(x_k) = y_k$ for all $k = 0, 1, 2, \dots, n$ that is $P(x)$ interpolates y_k at x_k for all $k = 0, 1, 2, \dots, n$.

x	x_0	x_1
y	y_0	y_1

There are two points

Degree of polynomial is 1

$$P(x_0) = y_0 \quad P(x) = a_0 + a_1x$$

$$P(x_1) = y_1$$

x	x_0	x_1	x_2
y	y_0	y_1	y_2

There are 3 points.

Degree of polynomial is 2.

$$P(x) = a_0 + a_1x + a_2x^2$$

$$P(x_0) = y_0$$

$$P(x_1) = y_1$$

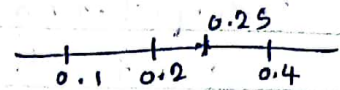
$$P(x_2) = y_2$$

Suppose.

$$x_0 = 0.1$$

$$x_1 = 0.2$$

$$x_2 = 0.4$$



1. Direct method :-

x	x_0	x_1
y	y_0	y_1

There are two points.

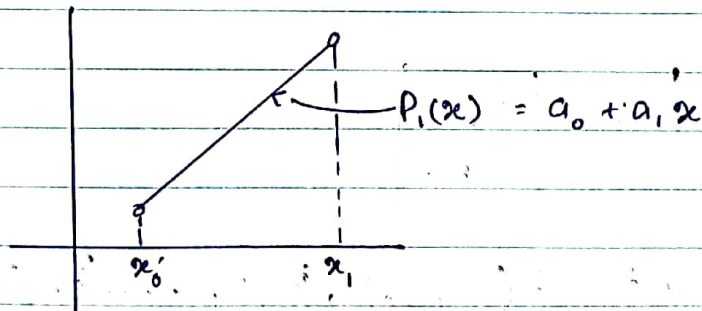
$\therefore$ Let $P_1(x) = a_0 + a_1x$ — Interpolating polynomial.
we must determine a_0 and a_1 .

$$P(x_0) = a_0 + a_1x_0 = y_0$$

$$P(x_1) = a_0 + a_1x_1 = y_1$$

Vandermonde matrix.

$$\begin{bmatrix} 1 & x_0 \\ 1 & x_1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \end{bmatrix} \leftarrow \text{solve to get } a_0 \text{ and } a_1$$



(ii)

x	x_0	x_1	x_2
y	y_0	y_1	y_2

There are three points.

Interpolating polynomial is $P_2(x) = a_0 + a_1x + a_2x^2$
 we must determine a_0, a_1, a_2 . Such that

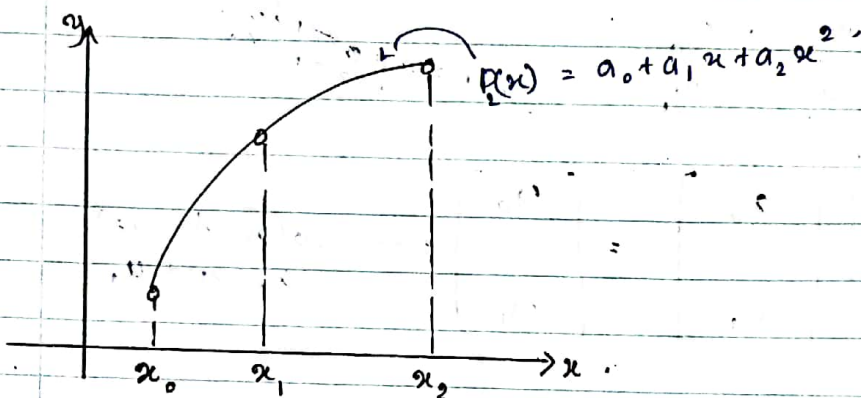
$$P_2(x_0) = y_0 \Rightarrow a_0 + a_1x_0 + a_2x_0^2 = y_0$$

$$P_2(x_1) = y_1 \Rightarrow a_0 + a_1x_1 + a_2x_1^2 = y_1$$

$$P_2(x_2) = y_2 \Rightarrow a_0 + a_1x_2 + a_2x_2^2 = y_2 \dots$$

$$\begin{bmatrix} 1 & x_0 & x_0^2 \\ 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \end{bmatrix} \leftarrow \text{solve this to get } a_0, a_1, a_2.$$

↑
vandermonde matrix.



(iii)

x	x_0	x_1	x_2	$\dots$	x_n
y	y_0	y_1	y_2	$\dots$	y_n

There are $n+1$ points : $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$

Let $P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ $\leftarrow$ Interpolation Polynomial

we must determine $a_0, a_1, a_2, \dots, a_n$ Such that

$$P(x_0) = y_0$$

$$P(x_1) = y_1$$

$$P(x_2) = y_2$$

$$\vdots$$

$$P(x_n) = y_n$$

Then we have ;

$$a_0 + a_1 x_0 + a_2 x_0^2 + a_3 x_0^3 + \dots + a_{n-1} x_0^{n-1} + a_n x_0^n = y_0$$

$$a_0 + a_1 x_1 + a_2 x_1^2 + a_3 x_1^3 + \dots + a_{n-1} x_1^{n-1} + a_n x_1^n = y_1$$

$$a_0 + a_1 x_2 + a_2 x_2^2 + a_3 x_2^3 + \dots + a_{n-1} x_2^{n-1} + a_n x_2^n = y_2$$

.

.

.

$$a_0 + a_1 x_n + a_2 x_n^2 + a_3 x_n^3 + \dots + a_{n-1} x_n^{n-1} + a_n x_n^n = y_n$$

$$\begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & x_2^2 & \dots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

↑
Vandermonde matrix

Solving this Vandermonde matrix, we can find $a_0, a_1, a_2, \dots, a_n$.

However, for large n , this direct method is not practical (not efficient; the Vandermonde matrix is ill-conditioned).

It is possible to find the polynomial in other forms that may be easier to use. Here we consider the Lagrange and Newton forms:

- Lagrange Interpolating polynomials -

~~Newton's Interpolating Polynomial.~~

~~Let (x_0, y_0) and (x_1, y_1) be two given points.
The first order Newton's polynomial is given~~

$$~~P_1(x) = a_0 + a_1(x - x_1)~~$$

x_0	x_1
y_0	y_1

* Let (x_0, y_0) and (x_1, y_1) be two given points.
The first order Lagrange polynomial $P_1(x)$ is given by

$$P_1(x) = a_0(x - x_1) + a_1(x - x_0)$$

x	x_0	x_1
y	y_0	y_1

we must determine a_0 and a_1

$$P_1(x_0) = a_0(x_0 - x_1) + a_1(x_0 - x_0) = y_0$$

$$\Rightarrow a_0 = \frac{y_0}{x_0 - x_1}$$

$$P_1(x_1) = a_0(x_1 - x_1) + a_1(x_1 - x_0) = y_1$$

$$\Rightarrow a_1 = \frac{y_1}{x_1 - x_0}$$

Here $x_0 \neq x_1$, since we consider distinct points.

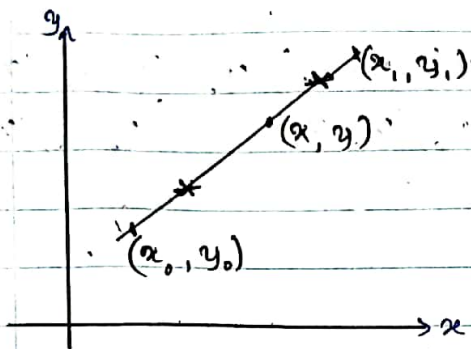
$$P_1(x) = \frac{(x-x_1)y_0}{(x_0-x_1)} + \frac{(x-x_0)y_1}{(x_1-x_0)}$$

where; $L_0(x)y_0 + L_1(x)y_1$

where; $L_0(x) = \frac{x-x_1}{x_0-x_1}$ and $L_1(x) = \frac{x-x_0}{x_1-x_0}$

Note:- $L_0(x_0) = \frac{x_0-x_1}{x_0-x_1} = 1$ $L_0(x_1) = \frac{x_1-x_1}{x_0-x_1} = 0$

$L_1(x_0) = 0$ and $L_1(x_1) = 1$



For three distinct points (x_0, y_0) , (x_1, y_1) and (x_2, y_2) , the second-order Lagrange polynomial is given by

$$P_2(x) = \underbrace{a_0(x-x_1)(x-x_2)}_{\hat{x}_0 \text{ absent}} + \underbrace{a_1(x-x_0)(x-x_2)}_{\hat{x}_1 \text{ absent}} + \underbrace{a_2(x-x_0)(x-x_1)}_{\hat{x}_2 \text{ absent}}$$

we must determine a_0 , a_1 and a_2

x	x_0	x_1	x_2
y	y_0	y_1	y_2

①

②

③

Also

$$P_2(x_0) = a_0 (x_0 - x_1)(x_0 - x_2) = y_0$$

$$a_0 = \frac{y_0}{(x_0 - x_1)(x_0 - x_2)}$$

$$P_2(x_1) = a_1 (x_1 - x_0)(x_1 - x_2) = y_1$$

$$a_1 = \frac{y_1}{(x_1 - x_0)(x_1 - x_2)}$$

$$P_2(x_2) = a_2 (x_2 - x_0)(x_2 - x_1) = y_2$$

$$\therefore a_2 = \frac{y_2}{(x_2 - x_0)(x_2 - x_1)}$$

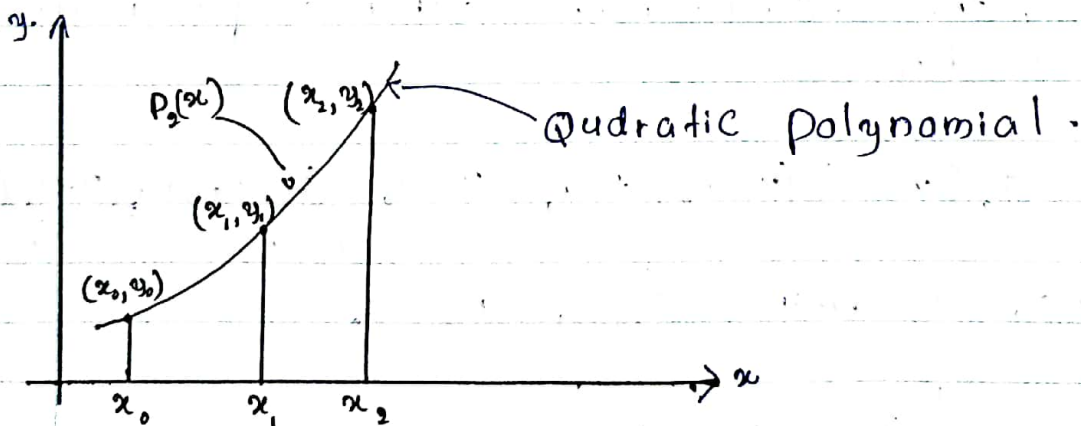
$$\begin{aligned} \therefore P_2(x) &= \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2 \\ &= L_0(x) y_0 + L_1(x) y_1 + L_2(x) y_2 \end{aligned}$$

Note:-

$$L_0(x_0) = 1, L_0(x_1) = 0, L_0(x_2) = 0$$

$$L_1(x_0) = 0, L_1(x_1) = 1, L_1(x_2) = 0$$

$$L_2(x_0) = 0, L_2(x_1) = 0, L_2(x_2) = 1$$



Now Consider $(n+1)$ distinct points which are given by $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

x	x_0	x_1	x_2	$\dots$	x_{n-1}	x_n
y	y_0	y_1	y_2	$\dots$	y_{n-1}	y_n

Then an n^{th} order Lagrange polynomial is defined by

$$P_n(x) = a_0(x-x_1)(x-x_2)\dots(x-x_n) + a_1(x-x_0)(x-x_2)\dots(x-x_n) + \dots + a_{n-1}(x-x_0)(x-x_1)\dots(x-x_n) + a_n(x-x_0)(x-x_1)\dots(x-x_{n-1})$$

Then we have ;

$$a_0 = \frac{y_0}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)}$$

$$= \frac{y_0}{\prod_{k=1}^n (x_0-x_k)}$$

$$a_1 = \frac{y_1}{(x_1-x_0)(x_1-x_2)\dots(x_1-x_n)}$$

$$= \frac{y_1}{\prod_{\substack{k=0 \\ k \neq 1}}^n (x_1-x_k)}$$

$\vdots$

$\vdots$

Ans

$$\sum_{k=1}^n a_k = a_1 + a_2 + \dots + a_n$$

$$\prod_{k=1}^n a_k = a_1 \times a_2 \times \dots \times a_n = a_1 a_2 a_3 \dots a_n$$

$$a_n = \frac{y_n}{(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})}$$

$$= \frac{y_n}{\prod_{k=0}^{n-1} (x_n - x_k)}$$

$$\therefore P_n(x) = \frac{(x - x_1)(x - x_2) \dots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \dots (x_0 - x_n)} y_0 + \frac{(x - x_0)(x - x_2) \dots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \dots (x_1 - x_n)} y_1$$

$$+ \dots + \frac{(x - x_0)(x - x_1) \dots (x - x_{n-1})}{(x_n - x_0)(x_n - x_1) \dots (x_n - x_{n-1})} y_n$$

$$= L_0(x) y_0 + L_1(x) y_1 + \dots + L_n(x) y_n$$

where

$$L_k(x) = \frac{(x - x_0)(x - x_1) \dots + (x - x_{k-1}) + (x - x_{k+1}) \dots (x - x_n)}{(x_k - x_0)(x_k - x_1) \dots (x_k - x_{k-1})(x_k - x_{k+1}) \dots (x_k - x_n)}$$

$$= \frac{\prod_{\substack{i=0 \\ i \neq k}}^n (x - x_i)}{\prod_{\substack{i=0 \\ i \neq k}}^n (x_k - x_i)} = \begin{cases} 1 & \text{if } i = k \\ 0 & \text{if } i \neq k \end{cases}$$

$$\therefore P_n(x) = \sum_{k=0}^n y_k L_k(x)$$

$$= \sum_{k=0}^n y_k \prod_{\substack{i=0 \\ i \neq k}}^n \frac{(x - x_i)}{(x_k - x_i)}$$

Algebra

Example :-

Find the linear interpolation polynomial that agrees with the data $f(0.1) = 0.1003$
 $f(0.4) = 0.4228$

x	0.1 x_0	0.4 x_1
y	0.1003 y_0	0.4228 y_1

$$P_1(x) = \frac{(x - x_1)}{x_0 - x_1} y_0 + \frac{(x - x_0)}{(x_1 - x_0)} y_1$$

$$= \frac{(x - 0.4)}{(0.1 - 0.4)} 0.1003 + \frac{(x - 0.1)}{(0.4 - 0.1)} 0.4228$$

$$P_1(x) = \frac{0.1003}{-0.3} (x - 0.4) + \frac{0.4228}{0.3} (x - 0.1)$$

Find $f(0.3)$:

$$f(0.3) \approx P_1(0.3) = \frac{0.1003}{-0.3} (0.3 - 0.4) + \frac{0.4228}{0.3} (0.3 - 0.1)$$

$$= \frac{0.1003}{-0.3} (0.3 - 0.4) + \frac{0.4228}{0.3} (0.3 - 0.1)$$

$$= 0.3153$$

Example 2:-

Find the quadratic interpolation polynomial agreeing with the data $f(0.1) = 0.1003$,
 $f(0.4) = 0.4228$ and $f(0.8) = 1.0296$.

x	0.1	0.4	0.8
y	0.1003	0.4228	1.0296

$$\begin{aligned}
 P_2(x) &= \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} y_0 + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} y_1 + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} y_2 \\
 &= \frac{(x-0.4)(x-0.8)}{(0.1-0.4)(0.1-0.8)} \times 0.1003 + \frac{(x-0.1)(x-0.8)}{(0.4-0.1)(0.4-0.8)} \times 0.4228 \\
 &\quad + \frac{(x-0.1)(x-0.4)}{(0.8-0.1)(0.8-0.4)} \times 1.0296
 \end{aligned}$$

Find $f(0.45)$

$$f(0.45) \approx P_2(0.45) =$$

Home work:- Consider the following data set

x	1	2	4	5	7
y	52	5	-5	-40	10

- (i) determine 4th order Lagrange polynomial that passes through the given points.
- ii) use the polynomial obtained in part i to determine the interpolated value for $x = 3$.
develop a matlab user define function that interpolates using a Lagrange Polynomial.
~~that is~~